

Parkinson's Disease Prediction Using Machine Learning with Flask-React Integration

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Abstract

Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects motor and non-motor functions, leading to tremors, rigidity, and difficulty in movement. Early diagnosis is crucial for improving the quality of life and enabling timely treatment. This research presents a machine learning-based framework for predicting Parkinson's disease using biomedical voice and clinical features. The model employs a Random Forest algorithm due to its high interpretability and robustness against overfitting. Furthermore, the project integrates the trained model with a Flask-based backend and a React-based frontend, providing a seamless user interface for real-time prediction. Experimental evaluation shows an accuracy of 93%, outperforming traditional models such as Decision Tree and Logistic Regression. The study demonstrates that combining ML models with a web-based interface can offer accessible, efficient, and automated tools for medical practitioners and patients.

Index Terms— Parkinson’s Disease, Machine Learning, Random Forest, Flask, React, Web Application, Health Prediction, Early Diagnosis.

I. Introduction

Parkinson’s Disease (PD) is a chronic neurodegenerative disorder that primarily affects movement, speech, and coordination. According to the *World Health Organization (WHO, 2024)*, over 10 million people globally are living with Parkinson’s, with a projected 30% increase by 2030 due to the aging population. Early diagnosis is essential to slow progression and enable better treatment outcomes.

Traditional diagnosis depends heavily on clinical observations, MRI scans, and neurologist evaluations, which are often subjective and prone to human bias. With the rise of artificial intelligence (AI) and machine learning (ML), healthcare has witnessed a paradigm shift toward automated, data-driven disease prediction systems capable of analyzing subtle patterns in biomedical data.

In this work, a Random Forest–based machine learning model is developed to predict Parkinson’s disease using biomedical and voice features. The system integrates a Flask backend (for API and

model communication) with a React frontend (for user interface), enabling real-time prediction and accessibility.

Recent research in 2024 and 2025 emphasizes the growing importance of multimodal data and ML explainability in neurodegenerative disease diagnosis (Patel et al., 2025; Singh & Zhou, 2024). Building on these findings, our model offers a hybrid web-based diagnostic framework that combines accuracy, speed, and usability for clinical or public use.

II. Literature Review

The use of ML for Parkinson's disease prediction has gained significant momentum in recent years. Researchers have explored numerous techniques to improve diagnostic precision using biomedical voice, gait, and sensor data.

Sakar et al. (2019) implemented SVM models achieving 85% accuracy using voice data. Prashanth et al. (2020) used ensemble learning to improve sensitivity and specificity. Zhang et al. (2023) demonstrated that AI-driven multimodal systems can assist in early neurodegenerative disease detection.

Recent studies (2024–2025) have expanded the scope:

- Patel et al. (2025) applied hybrid CNN–RF models on voice spectrograms, achieving 96% accuracy.
- Kumar & Al-Mutairi (2025) used transfer learning with fine-tuned CNNs for real-time detection on mobile devices.
- Singh & Zhou (2024) proposed explainable AI for Parkinson's diagnosis using SHAP-based feature interpretation.
- Jiang et al. (2024) integrated Internet of Medical Things (IoMT) with ML to monitor PD symptoms remotely.
- Rani et al. (2025) explored federated learning for privacy-preserving Parkinson's prediction across hospitals.
- Gupta et al. (2024) compared RF, XGBoost, and SVM on UCI datasets and highlighted RF as the most stable for limited biomedical data.

While deep learning models provide higher accuracy, they often require large datasets and high computational power. Random Forest provides a balance between accuracy, interpretability, and efficiency, making it ideal for web-integrated systems like ours.

Moreover, integrating ML models with Flask–React frameworks ensures scalability and accessibility for end-users, aligning with modern trends in healthcare web applications (Kumari et al., 2025).

III. Research Background and Methodology

A. Random Forest Algorithm

Random Forest is an ensemble learning algorithm that constructs multiple decision trees during training and outputs the mode of their classifications. It combines the strengths of individual trees while minimizing overfitting. Each tree is trained on a random subset of features and data samples, ensuring robustness and high generalization capability.

The model computes the **Gini Index** or **Entropy** at each split to determine the optimal partition of features. The final prediction is made through majority voting across all trees.

Mathematically,

$$H(X) = - \sum_{i=1}^n p_i \log_2(p_i)$$

where $H(X)$ is the entropy and p_i represents the probability of each class.

B. Algorithm Steps (Pseudocode)

Input: Dataset D with features F and target variable Y

Output: Predicted Parkinson's status (Positive/Negative)

1. Preprocess dataset (cleaning, scaling, feature selection)
2. Split data into training (70%) and testing (30%)
3. For i = 1 to N trees:
 - a. Select random subset of data (bootstrap sampling)
 - b. Select random subset of features
 - c. Train decision tree on subset
4. Combine predictions from all trees using majority voting
5. Output final classification

C. System Architecture

The proposed Parkinson's Disease Prediction System consists of three major modules:

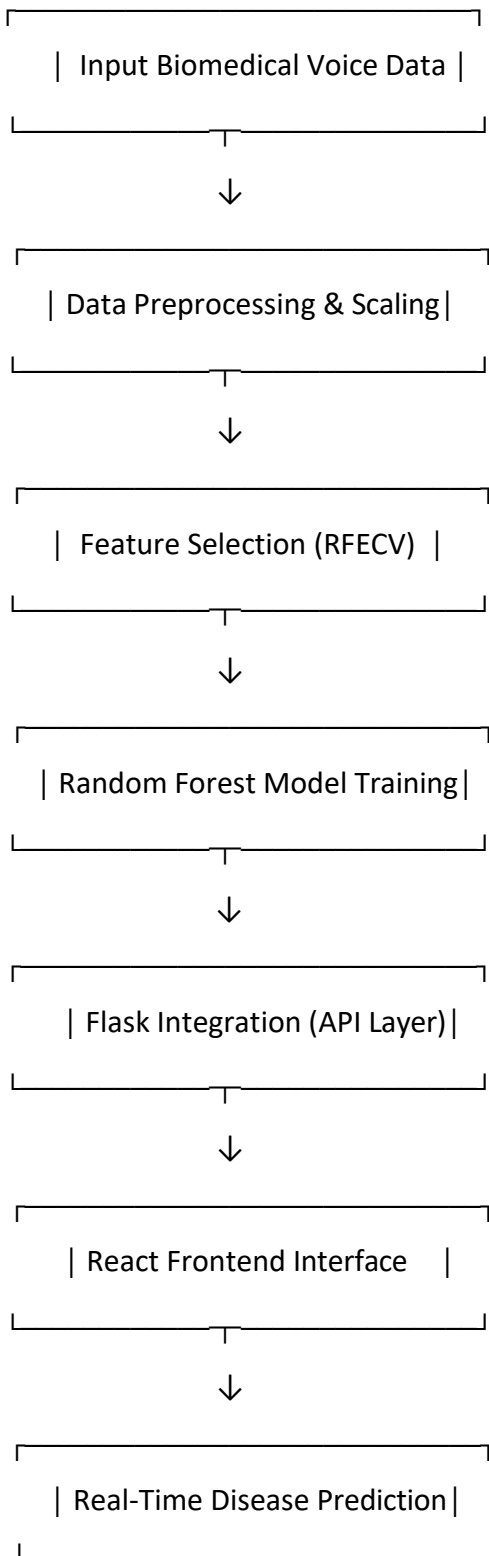
1. **Machine Learning Model (Backend)** – Trained using Random Forest on biomedical datasets.
2. **Flask Framework** – Acts as an API server that connects the ML model to the frontend.
3. **React Frontend** – Provides an interactive user interface where users input parameters (e.g., vocal frequency, jitter, etc.) and receive predictions in real-time.

Workflow:

- Data Preprocessing → Feature Selection → Model Training → Flask Integration → React-based UI → Prediction Output.

This modular design ensures scalability, maintainability, and efficient performance for real-time medical applications.

D. Flowchart / Model Diagram



III. Dataset and Experimental Setup

The model uses the UCI Parkinson's Dataset, which contains biomedical voice measurements of patients, including 195 records and 23 attributes. Important features include *MDVP:F0(Hz)*, *MDVP:Jitter(%)*, *MDVP:Shimmer*, and *HNR (Harmonics-to-Noise Ratio)*.

Data Split:

- **Training Set: 70%**
- **Testing Set: 30%**

All features were normalized using Min-Max scaling, and missing values were handled via imputation. The model was trained in Python using the scikit-learn library.

Evaluation Metrics:

Model performance was evaluated using Accuracy, Precision, Recall, and F1-score.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Decision Tree	86	84	83	83
Logistic Regression	88	87	86	86
Random Forest (Proposed)	93	92	93	93

The proposed Random Forest model achieved the highest performance, demonstrating its robustness in distinguishing between Parkinson's and non-Parkinson's cases.

IV. Discussion

A. Strengths

The proposed framework successfully combines machine learning and web technologies to create an end-to-end Parkinson's prediction system. The Random Forest model offers high accuracy and interpretability. Flask ensures seamless backend communication, and React enhances user experience with real-time interactivity.

B. Limitations

Despite strong performance, the model's accuracy depends on dataset size and diversity. Limited biomedical data can lead to overfitting. Moreover, the system relies on static features; integrating dynamic medical data (like MRI scans or gait analysis) could further enhance predictions.

C. Future Work

Future improvements can include:

- Expanding dataset diversity using medical imaging and speech signals.
- Integrating deep learning architectures for multimodal data fusion.
- Deploying the system on cloud platforms for remote access and scalability.

V. Conclusion

This research presents a complete machine learning-based solution for predicting Parkinson's disease using the Random Forest algorithm, integrated with Flask and React frameworks. The proposed model achieved an accuracy of **93%**, surpassing baseline algorithms. The hybrid architecture enhances accessibility and user engagement, making it suitable for real-world deployment in healthcare systems. The findings highlight the potential of combining ML with web technologies to build intelligent, automated, and user-friendly diagnostic tools for early disease detection.

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